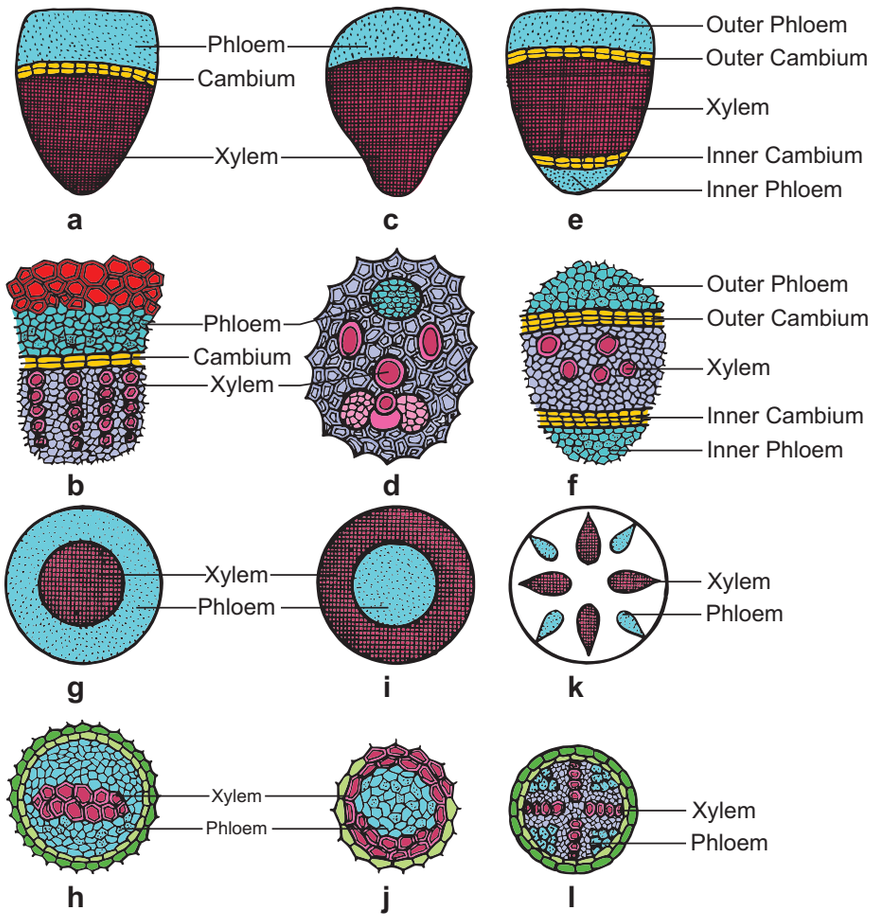


Figure 9.15: Types of vascular bundles

(a) and **(b)** - Conjoint, collateral and open; **(c)** and **(d)** - Conjoint, collateral and closed **(e)** and **(f)** - Conjoint, bicollateral and open; **(g)** and **(h)** - Concentric and amphicribral; **(i)** and **(j)** - Concentric and amphivasal; **(k)** and **(l)** - Radial

9.7 Comparison of Primary Structure – Dicot and Monocot Root, Stem and Leaf

Primary Structure of Dicot Root – Bean Root

The transverse section of the dicot root (Bean) shows the following plan of arrangement of tissues from the periphery to the centre.

Piliferous Layer or Epiblema

The outermost layer of the root is called **piliferous layer or epiblema.** It is made up of single layer of parenchyma cells which are arranged compactly without intercellular spaces. It is devoid of epidermal pores and cuticle. It possesses root hairs which are single celled. It absorbs water and mineral salts from the soil. The chief function of piliferous layer is **protection.**

Cortex

Cortex consists of only parenchyma cells. These cells are loosely arranged with intercellular spaces to make gaseous exchange easier. These cells may store food reserves. The cells are oval or rounded in shape. Sometimes they are polygonal due to mutual pressure. Though chloroplasts are absent in the cortical cells, starch grain are stored in them. The cells also possess leucoplasts. The innermost layer of the cortex is endodermis. Endodermis is made up of single layer of barrel shaped parenchymatous cells. Stele is completely surrounded by endodermis. The radial and the inner tangential walls of endodermal cells are thickened with **suberin and lignin.** This thickening was first noted by **Robert Casparay** in 1965. So these thickenings are called **casparian strips.** But these casparian

11th Botany_EM_Chapter 9.indd 209 06/01/2023 20:02:06